

# Production of Health

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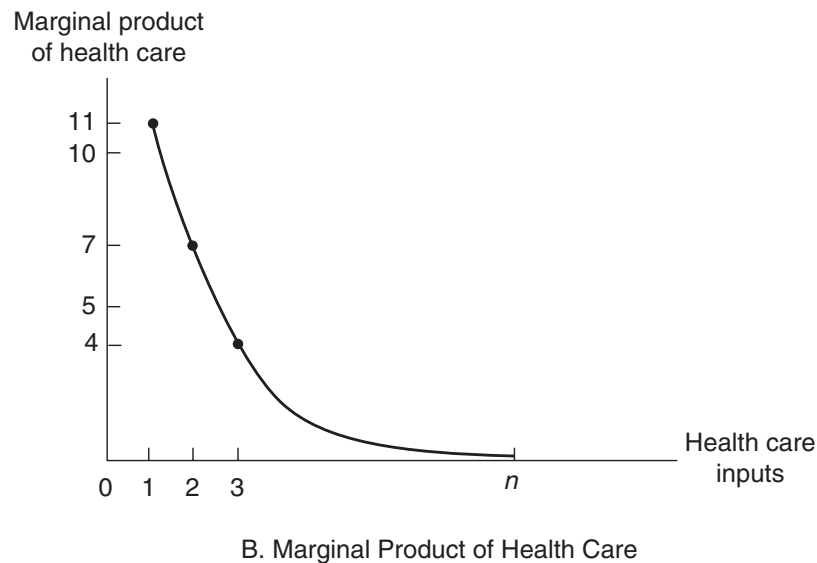
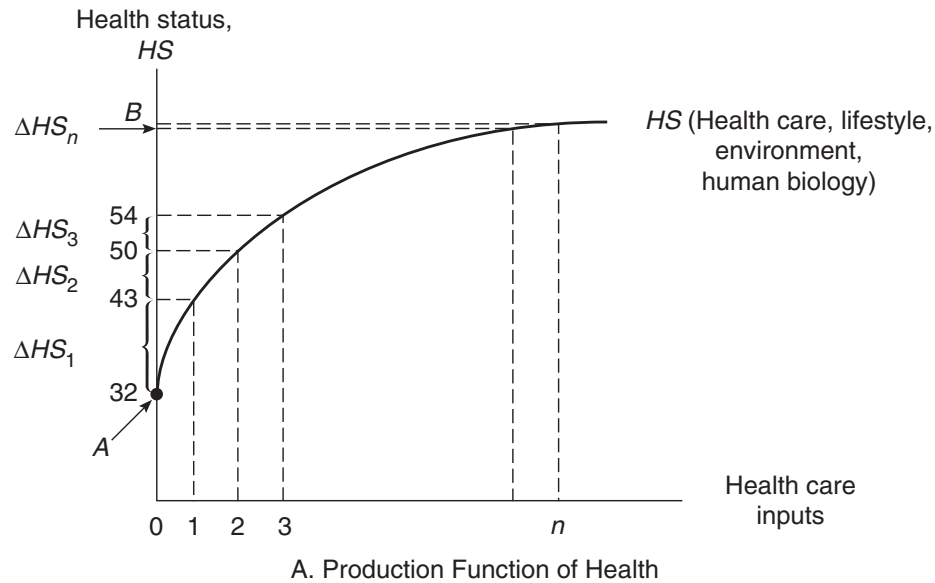
The production of health presents a central concern to the health economist and to public policy. Consider that the role of health care in society, including medical care provided by physicians, is ultimately a production question. We must learn about the determinants of health and about the contribution of health care. We can then better understand what decisions, both personal and public, will best produce health.

In medical terminology, this chapter addresses the efficacy and effectiveness of all those features of life, not only medical care, that plausibly contribute to our health. Unlike the typical doctor in practice, however, we look for evidence of the response of a “treatment” in the change in the health status of populations, as opposed to the treatment response of a medicine for the individual patient. We will see that the two approaches must remain in harmony and that both are fundamentally searches for causal relationships.

## THE PRODUCTION FUNCTION OF HEALTH

A production function summarizes the relationship between inputs and outputs. The study of the production of health function requires that we inquire about the relationship between health inputs and health. The answers that economists and medical historians offer to this question surprise many people. First, the contribution of practitioner-provided health care to the historical downward trends in population mortality rates was probably negligible at least until well into the twentieth century. Second, while the total contribution of health care is substantial in the modern day, its marginal contribution in some cases is small.

This distinction between total and marginal contributions is crucial to understanding these issues. To illustrate this distinction, consider Figure 5-1A, which exhibits a theoretical health status production function for the population. Set aside the difficulties of measuring health status in populations, and assume that we have defined an adequate health status (*HS*) measure. Health status here is an increasing function of health care. Also, to

**FIGURE 5-1** Production of Health

avoid a perspective that is too narrowly focused on health care, we specify further that health status depends at least upon the population's biological endowment, environment, and lifestyle.<sup>1</sup> Thus,  $HS = HS$  (Health Care, Lifestyle, Environment, Human Biology). Improvements in any of these latter three factors will shift the curve upward.

A production function describes the relationship of flows of inputs and flows of outputs over a specified time period, so the inputs and output in Figure 5-1A are measured over an implied period, such as a year. In practice, we might use the number of healthy days experienced by the population per capita, mortality rates, or disability days, to indicate health status.

<sup>1</sup> This categorization of groups of inputs is not the only reasonable one, but it illustrates the main issues, and it has an excellent history. It was developed by the Canadian government for its pioneering work begun almost four decades ago (Lalonde, 1974).

To simplify the depiction, we have reduced all health care inputs into one scale called Health Care. In reality, health care consists of many health care inputs. Some of them include medical care provided by doctors of medicine or osteopathy, but other health care professionals also provide care. Conceptually, the health care measure,  $HC$ , may be thought of as an aggregate of all these types of health care, the aggregation being based on dollar values.

The marginal contribution of health care is its marginal product, meaning the increment to health caused by one extra unit of Health Care, holding all other inputs constant. Increasing Health Care from zero to one unit in Figure 5-1A improves health status by  $\Delta HS_1$ , the first unit's marginal product. Numerically, this first unit of Health Care has increased the health status index from 32 to 43;  $\Delta HS_1 = 11$  Health Status units. The next unit of medical care delivers a marginal product of  $\Delta HS_2 = 7$ , and so on.

These marginal products are diminishing in size, illustrating the law of diminishing marginal returns. If society employs a total of  $n$  units of Health Care, then the total contribution of Health Care is the sum of the marginal products of each of the  $n$  units. This total contribution as shown,  $AB$ , may be substantial. However, the marginal product of the  $n^{\text{th}}$  unit of medical care is  $\Delta HS_n$ , and it is small. In fact, we are nearly on the "flat of the curve." Marginal product is graphed on Figure 5-1B.

We have drawn the health production function as a rising curve that flattens out at higher levels of health care but never bends downward. Would the health production function eventually bend downward? Is it possible to get too much health care so that the health of the population is harmed? This is a logical possibility under at least two scenarios. *Iatrogenic* (meaning provider-caused) disease is an inevitable by-product of many medical interventions. For example, each surgery has its risks. Combinations of drugs may have unforeseen and adverse interactions. If the rate of iatrogenic disease does not fall while diminishing returns sets in, it is possible for the balance of help and harm from health care to be a net harm.

Medical scientists, such as Cochrane (1972), have pressed the case that much medical care as often practiced has only weak scientific basis, making iatrogenesis a real probability. Writing for the public audience, Dubos (1960) and Illich (1976) once warned of a medical "nemesis" taking away our abilities to face the natural hardships of life by "medicalizing" these problems. Illich argued that this medicalization would lead to less personal effort to preserve health and less personal determination to persevere; the result becomes a decline in the health of the population and thus a negative marginal product for medical care.<sup>2</sup>

Return to the distinction between total product and marginal product. Often, the marginals, rather than the totals, are relevant to policy propositions. For example, no one seriously recommends that society eliminate all health care spending. However, it is reasonable to ask whether society would be better off if it could reduce health care expenditures by \$1 billion and invest those funds in another productive use, such as housing, education, transportation, defense, or other consumption. We could even reasonably ask if health itself could be improved by transferring the marginal \$1 billion to environmental or lifestyle improvements.

Many of our government programs encourage health care use in certain population groups, such as the poor and elderly. Other programs, such as tax preferences for health insurance, provide benefits for those who are neither poor nor elderly and encourage their consumption of health care. The theoretical issues raised here suggest that we question the wisdom of each of our programs. The theoretical questions can be investigated with data of several kinds either directly or indirectly relevant to the production of health issue. We begin with the historical role of medicine, which indirectly bears on the issue of health production. After providing an overview of these efforts, largely the work of medical and economic historians, we then turn to econometric studies of the modern-day production function.

<sup>2</sup> The medicalization argument was reinforced by Thomas McKeown's (1976) research showing that medical care warranted little credit for the historical declines in mortality rates in England and Wales. The works by McKeown and by Cochrane remain the reference points for the continuing debate within medicine and the social sciences over the role of medicine (Alvarez-Dardet and Ruiz, 1993; Farmer and Nardell, 1998; Frenk, 1998).

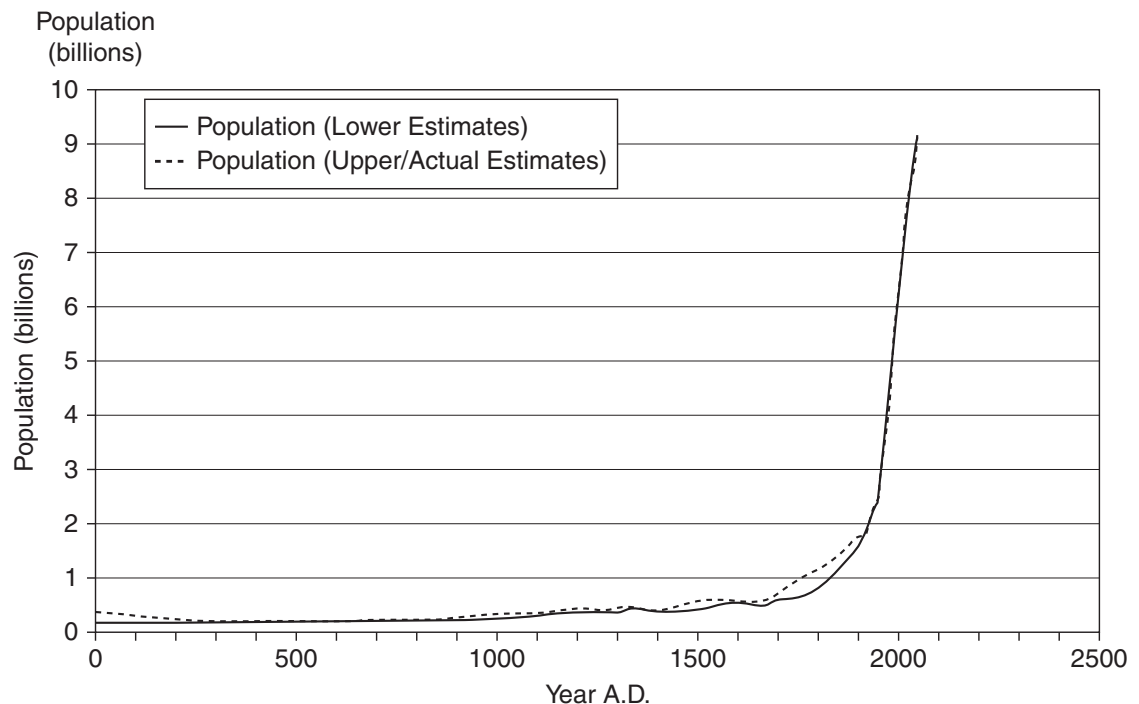
## THE HISTORICAL ROLE OF MEDICINE AND HEALTH CARE

Many medical historians agree that practitioner-provided medical interventions played only a small, perhaps negligible, role in the historical decline in population mortality rates. Effective medicine is a fairly recent phenomenon, and the delivery of effective medical interventions on a scale sufficient to affect population health indicators most likely appeared only well into the twentieth century. Though the magnitudes of other causes of mortality declines are still disputed, it is clear that a larger role, one of the most significant ones, might be attributed to public health measures and the spread of knowledge of the sources of disease. However, a number of scholars in this field attribute the largest share of the credit to improvements in environment, particularly to the greatly increased supply of foodstuffs that became available due to the agricultural and industrial revolutions.

### The Rising Population and the Role of Medicine

The notion that medicine played a relatively minor historical role is certainly not new, and it has been asserted by researchers of various ideologies. This point of view is associated with the work of Thomas McKeown (1976), who focused on the dramatic rise in population in England and Wales from 1750 to the modern day.

The pattern of world population growth, including population growth in England and Wales, has interested many scholars, including McKeown. World population is hard to estimate for the distant past, but research by the United Nations (1996) and others show that something extraordinary happened during the last 300 years. In the first century the population was roughly 300 million. For a thousand years thereafter, until the era of Viking ships, little or no change occurred. By the Age of Enlightenment, starting just before 1700, the population may have risen to 600 million. Then things began to change rapidly. Within a single century, the world population passed 1 billion people. The next 5 billion arrived within a mere 200 years. What had happened? Figure 5-2, based on United Nations data, reveals this startling pattern.



**FIGURE 5-2** World Population from 0 B.C.E. to the Modern Day Source: <http://www.ciese.org/curriculum/popgrowthproj/worldpop.html>. Data from U.S. Census Bureau.

Returning to the history of England and Wales, the large rise in their populations in the period following 1750 is to a large degree a story of the population's health. Population increase comes from increased birth rates, reduced mortality, or increased net in-migration. Migration was not an important source of population increase in England and Wales; when accurate birth rate and death rate data became available from 1841, these data alone proved able to account for the population change. Likewise, fertility probably did not account for the change because recorded birth rates have declined during the period since data have become available. Declines in birth rates are a common finding in countries undergoing industrialization and modernization. In contrast, recorded mortality rates did decline substantially.

McKeown began by investigating which diseases contributed to the decline in death rates. Mortality data are very limited prior to the mid-1800s, but the records revealed an emerging picture. Table 5-1 shows death rates by disease category for three time periods. The table shows that airborne infectious diseases account for the largest single portion of mortality reduction, and waterborne infectious diseases also make up a substantial portion of known causes. Regarding the airborne diseases, other data suggest that the main airborne diseases showing a decline in mortality include tuberculosis, bronchitis, pneumonia, and influenza.

### What Caused the Mortality Rate Declines? Was It Medicine?

Many presume that the declines in the mortality rates were due to improvements in medical science provided to the public through medical practice, but counterarguments to this proposition bring it into question. In most cases, an effective specific medical intervention was not available until late in the period, well after the greater part of the mortality decline had occurred.

The argument can be illustrated for the cases of respiratory tuberculosis and a group of three upper respiratory diseases—bronchitis, pneumonia, and influenza. Mortality rates for these diseases fell to relatively low levels prior to the availability of effective medical interventions, whose availability occurred respectively after 1930, and for some cases well into the 1950s and 1960s.

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**TABLE 5-1 Death Rates\* (per million) in 1848–1854, 1901, and 1907**

|                                                   | 1848–1854 | 1901   | 1971  | Percentage of Reduction<br>(1848–1854 to 1971)<br>Attributable to Each<br>Category | For Each Category,<br>Percentage of<br>Reduction<br>(1848–1854 to 1971)<br>That Occurred<br>Before 1901 |
|---------------------------------------------------|-----------|--------|-------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| I. Conditions attributable to microorganisms:     |           |        |       |                                                                                    |                                                                                                         |
| 1. Airborne diseases                              | 7,259     | 5,122  | 619   | 39                                                                                 | 32                                                                                                      |
| 2. Water- and foodborne diseases                  | 3,562     | 1,931  | 35    | 21                                                                                 | 46                                                                                                      |
| 3. Other conditions                               | 2,144     | 1,415  | 60    | 12                                                                                 | 35                                                                                                      |
| Total                                             | 12,965    | 8,468  | 714   | 72                                                                                 | 37                                                                                                      |
| II. Conditions not attributable to microorganisms |           |        |       |                                                                                    |                                                                                                         |
|                                                   | 8,891     | 8,490  | 4,070 | 28                                                                                 | 8                                                                                                       |
| All diseases                                      | 21,856    | 16,958 | 5,384 | 100                                                                                | 29                                                                                                      |

\*Standardized to the age/sex distribution of the 1901 population.

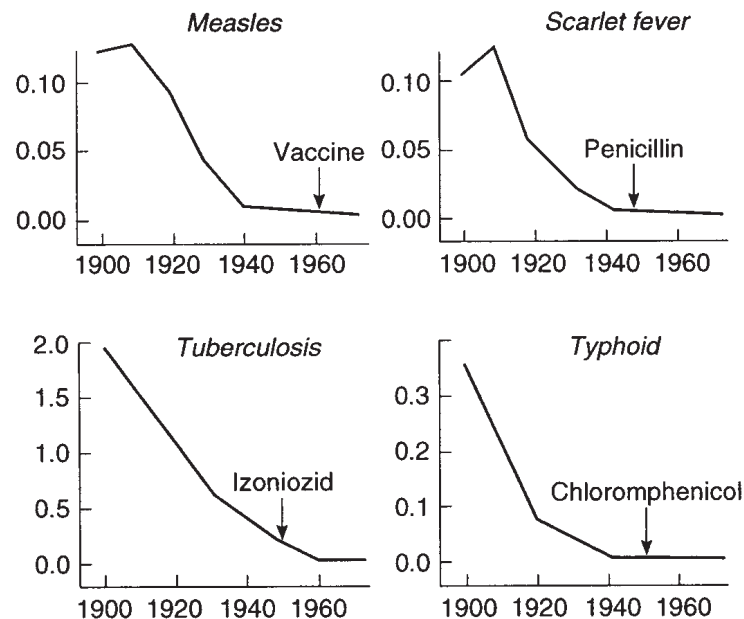
Source: reprinted from McKeown, Thomas, *The Modern Rise of Population*. New York: Academic Press. 1976 (p. 54).

**BOX 5-1****Tuberculosis and *The Magic Mountain***

Thanks to the efforts of writers such as Nobelist Thomas Mann, the tuberculosis (TB) sanitarium of 100 years ago has found a permanent place in literature. Mann's novel *The Magic Mountain* describes with a mastery of medical detail, often admired by physicians, the characters' struggle with the deadly disease. Mann's incomparable development of character and dialog bring a reality to the society unique to this sanatorium, representing one he had himself visited. Hans Castorp, a young German engineer, is smitten with Clavdia Chauchat, a young woman at the "good" Russian table. The Italian Settembrini's intellectual arguments with Naptha can result only in a duel. We grieve when Hans's cousin Joachim, a good, simple military man, succumbs to TB and his body is sledded down the mountain. Hans survives to return to the lands below.

Though effective chemical interventions were not available until after 1940, doctors did treat TB prior to 1940, notably with the widespread use of sanatoria such as the one in *The Magic Mountain*. Declines in TB mortality during the period studied by McKeown represented perhaps the most important example of declines in the mortality rate. Can we credit the sanatoria for this progress? This is unlikely, because it is unlikely that their capacity was ever large enough to affect the pattern of mortality in populations. For an account of the retreat of tuberculosis, see Smith (1988).

to relatively low levels prior to the availability of effective medical interventions, whose availability occurred respectively after 1930, and for some cases well into the 1950s and 1960s. The picture is shared by waterborne diseases. About 95 percent of the mortality declines in cholera, diarrhea, and dysentery occurred prior to the 1930s, when intravenous therapies became available. Likewise, typhoid and typhus mortality already had fallen to low levels by the beginning of the twentieth century. The pattern McKeown found for England and Wales also can be illustrated for the United States. McKinlay and McKinlay (1977) provided data for the United States from 1900 to 1973. Figure 5-3 shows these



**FIGURE 5-3** Fall in the Standardized Death Rate per 1,000 Population for Four Common Infectious Diseases in Relation to Specific Medical Measures for the United States Source: Reprinted from *Milbank Memorial Fund Quarterly/Health and Society*, John B. McKinlay and Sonja M. McKinlay, "The Questionable Contribution of Medical Measures to the Decline of Mortality in the United States in the Twentieth Century," *Milbank Memorial Fund Quarterly/Health and Society* 55 (1977): 405–428, with the permission of Blackwell Publishers.

patterns for several infectious diseases. In most cases, as is shown, the availability of the effective medical intervention occurs well after the majority of the mortality declines.

One of the most important changes in mortality in the twentieth century was the decline in infant mortality. Does this type of mortality follow the same pattern? A highly readable account of the modern historical pattern of infant mortality is offered in Victor Fuchs's *Who Shall Live?* (1975). Fuchs noted that infant mortality rates in New York City improved markedly from 1900 to 1930 and that this decline was due to declines in deaths from "pneumonia-diarrhea" complex. Fuchs concluded: "It is important to realize that medical care played almost no role in this decline. While we do not know the precise causes, it is believed that rising living standards, the spread of literacy and education, and a substantial fall in the birth rate all played a part" (p. 32).

Antimicrobial drugs were introduced in the 1930s. Between 1935 and 1950, the fall in infant death rates accelerated. Fuchs proposed that medical advances and rising living standards both contributed to the reduction in infant deaths during this period. Declines in infant deaths flattened somewhat beginning about 1950 but resumed a stronger decline about 1965. If specific effective curative medicines were not largely responsible for mortality declines, is it nevertheless possible that other tools in the physician's black bag were effective? Unfortunately, this too is unlikely. The problem is that there probably were few effective tools available until well into the twentieth century. Even a clear knowledge of what caused disease was not widespread until the 1900s.

**NUTRITION REDUCED MORTALITY** Two of the most respected students of the mortality decline, medical historian Thomas McKeown (1976) and economic historian Robert Fogel (2004), argued strongly that the main cause was improved nutrition. McKeown reasoned by process of elimination. As we have just seen, he showed the medicine interventions could not have been the cause, a claim that is still widely accepted. He considered other possibilities one by one. For example, some have suggested that perhaps the infectious organisms had spontaneously mutated and became harmless; he pointed out that the chances were remote that so many independent organisms had randomly mutated at about the same time.

McKeown also dismissed public health as a major cause, however, and this argument was to become controversial. If we re-examine his work in Table 5-1, we see that the largest portion of mortality decline from 1848 to 1971 was due to declines in mortality from airborne diseases. He argued that public health projects, which focused on improving water quality and the safety of food, could have little effect on airborne diseases. McKeown clearly understood that clean water and pasteurized milk were important to improved health, but he claimed that these benefits came late in the historical era of mortality declines. Supporting his claim about the timing of public health, consider that the role of germs was not understood until the mid-1800s, about the time that public health came into being, and pasteurization of milk did not start until around 1870 and its widespread commercial use did not come until well into the twentieth century. Having eliminated everything else, in his reasoning McKeown assumed that the great benefactor that transformed the developed countries from high mortality to low mortality must have been improved nutrition.

This argument for the primacy of nutrition, however, provided no direct evidence that nutrition improves health. Robert Fogel (2004) provided that needed evidence. He established that after the mid-eighteenth century, calorie intake of Europeans increased tremendously. At about the same time, their average height also increased substantially. The relationship of height to health is now well known; the Waaler Curve established that, for any given body mass, taller people (up to a point) have greater life expectancy (Fogel, 2004). We also now understand how better nutrition makes an individual better able to resist infectious disease. Fogel went on to study in great detail the heights and records of Civil War soldiers in the United States. His research led him to claim that nutrition played the major role in what the title of his recent book calls: *The Escape from Hunger and Premature Death, 1700–2100*.

**PUBLIC HEALTH REDUCED MORTALITY** Other historical analysts take issue with the proposition that nutrition was the main cause of the mortality reductions. The crux of the issue is when the era of mortality reductions began. Public health advocates claim, contrary to McKeown, that the major



declines did not start until around 1870, and if they began this late, then public health, which began about 1850, would have come in time to contribute. We know that the era from 1870 to about 1940 completed the “epidemiological transition.” This phrase describes the remarkable transition in developed countries from when infectious disease was the major cause of death to a time when it became of only minor importance to population health. It is instructive to examine what public health accomplished during this period.

By 1870, cities had grown rapidly without the planning and development we now consider essential to a healthy environment. During this era, urban centers eventually and painfully slowly overcame their status of having worse mortality rates and general health than the countryside, the “urban deficit.” Streets contained animal excrement, sewer systems were designed mainly for storm water, and water supplies were often delivered in lead pipes. The transition from water tainted with infectious organisms to clean water supplies was the most dramatic change in the health environments of city dwellers (see Box 5-2, “The Importance of Clean Water,” for this story).

To summarize, the period from 1750 to the present contains three strands of health-related phenomena: (1) growth in life expectancy; (2) improved nutrition; and (3) improved public health. The difficulty is how to sort out which relationships proved most important. We see the importance of nutrition to body mass and height, the keys to health in the Waaler Curve (Fogel, 2004). This is compelling evidence. However those who believe that public health was of key importance can point to the clean-up of cities, also compelling evidence. They point out, for example, that we know the modern Chinese people are not as tall as Americans, yet their life expectancies are much higher than would have been expected. Does not this point to the modern adoption of public health measures, which now can take place very rapidly? We will see shortly that these historical puzzles are not merely of “academic” interest but are critical for the growth in well-being of the lesser developed world.

## BOX 5-2

### The Importance of Clean Water

If transported by time machine back to the mid-nineteenth century, you would find it difficult to survive. This is because your modern body mass and height could barely be sustained by the small average quantities of available calories. But if you did survive and went to live in a city, you would find that in your weakened condition you would be very susceptible to infectious disease organisms permeating your environment, and especially in the water.

Even in 1900, waterborne infectious disease accounted for one-quarter of the deaths from infectious disease. Public health campaigns, which were painfully slow in gaining acceptance, cleaned up the water. They introduced the filtering of city water through sand. They fought to have sewage discharged at a safe distance from water intakes. In prior cases, cities had discharged waste directly into the same lakes or streams that provided drinking water. Water closets were introduced in about 1870, and these discharged human waste into a city sewer system that often could not handle it and overflowed even into the streets. Public health also introduced chlorination of the water supplies. If the earlier contaminating practices seem obvious and foolish to us, we need to remember that germ theory had only recently arrived, and pasteurization was discovered only in the late 1800s.

Cutler and Miller (2005) estimate that filtration of city water brought reductions in total mortality of 16 percent, and reduced infant mortality by 43 percent, in the 12 American cities studied. Applying cost-benefit principles, the researchers found that the ratio of benefits to costs in the filtration projects was about 23 to 1. This is history to us, but it is present-day reality to less developed countries, where over 1 billion people lack access to clean water (Cutler and Miller 2005). The United Nations has declared the 2005–2015 period the International Decade for Action on Water. More on the U.N. program can be found on the Web by searching for “millennium development goals.”



### What Lessons Are Learned from the Medical Historian?

We cannot conclude that medical research is unimportant in history or in the present day. Medical research contributes not only through improvements to medical practice, but also through its influence on health-enhancing practices. Typhoid provides a good example. Mortality from typhoid declined substantially well before the arrival by 1950 of chloramphenicol. Medical research, however, contributed to our understanding of the cause and transmission of typhoid and generated public health measures such as filtering public water supplies, chlorination of water supplies, and establishment of drinking-water standards. All of these factors occurred historically in time to have a major effect on mortality. Selma Mushkin (1979) estimates that medical research accounted for almost one-third of the cost savings to society from reduction in sickness or death rates in the United States from 1900 to 1975. The period 1900 to 1930 accounts for half the value of medical research effects, even though it came largely before the specific effective medical practice interventions.

Investments in medical research play a major role in our health and well-being. Murphy and Topel (2005) used people's willingness to pay for advances in medical knowledge as a measure of its value. They estimated that the contribution of medical research to mortality reductions from 1970 to 2000 added \$3.2 trillion to national wealth. Since 1970, reductions in heart disease mortality alone have been worth about \$1.5 trillion. The medical research share of these gains, even if this amounted to only 10 percent of the total, easily compensates for the \$36 billion we invest in it annually.

Second, perhaps the best result of this overview is a healthy skepticism toward the effectiveness of any given medical practice, and more importantly, to its significance and benefit to the population. It is in this spirit that the U.S. government has increasingly come to fund outcome studies. Outcome studies seek to address the effectiveness and appropriateness of specific medical practices on patient outcomes. The studies attempt to reduce the prevalent uncertainties in medical practice, and they offer important inquiries into the wisdom of using the marginal billion dollars on medical care delivery, particularly in terms of costs and benefits to the population as a whole. In the same spirit, "evidence-based medicine" aims to close the gap between outcomes research and physician practice.

Finally, and most importantly, these historical puzzles have relevance to the progress and public investment practices of lesser developed countries, who have scarce resources with which to invest in either industrial growth or to invest directly into health measures and public health improvements. If improved nutrition is the key to population health, then perhaps industrial growth will bring the best overall gains. If public health is the most direct and productive way to achieve health, it is also a route to develop a more productive workforce. These questions easily merit diligent research. We note also that there is also an immense quasi-natural experiment to observe. Fogel (2000) and others report that world poverty rates are dropping sharply; the number of people in extreme poverty, those living on only a \$1 a day, has dropped sharply in the last two decades. We will want to see population health status in the developing countries improving.

## THE PRODUCTION OF HEALTH IN THE MODERN DAY

The investigation of the modern health production function requires econometric techniques. An understanding of the strengths and limitations of these contributions requires attention to the underlying conceptual issues.

### Preliminary Issues

Two conceptual issues bear on our interpretation of the results. These two issues can be posed as questions faced by every researcher: (1) how to measure health, the dependent variable in these studies, and (2) how to eliminate biases in the estimates.

**HOW TO MEASURE HEALTH** Consider the measurement of health. We desire a measure of population health status that captures the aspects of health status that are meaningful and that we can measure with adequate precision. It is difficult to attain both of these goals. We are most confident in the accuracy of mortality rate data, but mortality rates do not adequately capture several meaningful aspects of health status, such as reduction in pain and suffering and other improvements in the quality of life. The approach of past research in this field is understandable. Researchers have emphasized mortality data because of their accuracy, as well as because of their importance in the public mind. However analysts have used other indicators of health status, such as morbidity (illness) rates and disability days. In discussing the empirical literature, we begin with the mortality studies and then consider studies of morbidity.

**ELIMINATING BIASES—REDUCED FORM VERSUS STRUCTURAL EQUATIONS** Consider that the statistical estimates often rely on *reduced form* equations, which are practical representations of the underlying true model of the phenomenon. The true model is based on what are called *structural* equations. Estimation based on the reduced form equations can lead to misinterpretations.<sup>3</sup>

### The Contribution of Health Care to Population Health: The Modern Era

Health economists inevitably use different study designs and data sources to estimate the marginal product of health care. The resulting numbers need to be converted to a common basis, and for this we use the elasticity of health with respect to expenditure on health care inputs:

$$E_{\text{Health:Expenditure}} = \frac{\% \text{ change in health}}{\% \text{ change in health care expenditures}}$$

Table 5-2 reports the elasticities from several studies of the production of health. Each study applies econometric methods to analyze survey data; these range from statewide data to data on county groups (Hadley, 1982, 1988) to data on individuals (Sickles and Yazbeck, 1998). Though not shown in the table, recent investigations (Cremieux, Oulette, and Pilon, 1999) confirm a significant contribution of health spending to reduce infant mortality (in Canada).

The several studies offer insights because of their differences. Some suggest that lifestyle and environment expenditures could provide more benefits per dollar of cost than health care; and efforts to improve schooling or reduce cigarette smoking offer appealing trade-offs with health care spending. The Hadley studies helped establish that health care spending makes a statistically significant contribution to health and argues that health care passes benefit-cost criteria at the margin. The latest study follows its subjects through time.

**TABLE 5-2 Measuring the Contribution of Health Care to Population Health**

| Study Cited         | Date of Study | Health Care Elasticity* | Significant? |
|---------------------|---------------|-------------------------|--------------|
| Hadley              | 1982          | 0.12 to 0.17            | Yes          |
| Hadley              | 1988          | 0.20 to 1.00            | Yes          |
| Sickles and Yazbeck | 1998          | 0.03 to 0.05            | Yes          |

\*This is the elasticity of health with respect to health care expenditure.

<sup>3</sup> For further discussion of this issue and method, see Rosenzweig and Schultz (1983); Grossman and Joyce (1990); and Atkinson and Crocker (1992).

A study with both quality and novelty of design adds support to our finding that medical expenditures improve health (Almond, et al. 2010). Low birth-weight babies have a worse chance of surviving, and with “very low birth weight” defined as below 1500 grams. It will seem odd, but babies just below that cutoff had one percent *lower* mortality rates than babies just above it. The reason was that “very low birth weight” qualifies the baby for special medical treatment (with extra costs). The authors put these facts together finding that the marginal product of the extra care significantly contributed to health. They further concluded that the cost of saving a statistical life of a newborn with birth weight near 1500 grams is on the order of \$550,000 in 2006 dollars (Almond et al., 2010). Based on value of life estimates (see Chapter 4), these medical efforts are clearly worth it.

### Is Health Care Worth It?

Are we, then, literally on the “flat of the curve”, getting zero marginal product from the marginal investment in health care? The answer is clearly no. Murphy and Topel (2005), as we have seen, estimate that the American gains in life expectancy from 1970 to 2000, based on willingness-to-pay principles, are worth over \$3 trillion. We invest a small fraction of that in health technology research. Much of the gain is probably due to health care. Similarly Cutler (2004) ties 40 percent of the gains in life expectancy between 1950 and 2000 to medical improvements in just two categories: low-weight infant care and cardiovascular disease treatment. Again, based on willingness to pay, the life gains are worth the extra costs of medical care overall—not even counting the benefits from other forms of health care.

Look again at the marginal benefit of health care (technology held constant), which is small but not zero. We earlier suggested that the health production elasticity of health care on the margin is about 0.10. To illustrate the meaning of this elasticity, suppose that 0.10 were the true production elasticity of health care, and let Congress reallocate \$250 billion from other programs to health care expenditure. In 2011, the United States spent \$2.5 trillion on health care. The \$250 billion transfer would increase health care expenditure by 10 percent. We would extrapolate the improvement in health to be  $10 \times 0.10 = 1$  percent. If we define “health” by average life expectancy, a 1-percent gain would mean an increased average life expectancy of  $78 \times 0.01 = 0.78$  year, provided this increased expenditure continues indefinitely. Spread over the population (for a life span), however, the 0.78 year of life could lead to incremental benefits that exceed the incremental costs.

#### BOX 5-3

##### Sulfa: A Drug That Really Changed Things

At a time when cost-control planners seek to sort out medicine that is expensive but not very effective, sulfa is a good example to remind us that there are medicines and practices that work extraordinarily well. It was the miracle drug of the 1930s, made available in 1937 several years before penicillin appeared. Sulfa was a major player in the epidemiological transition from a time when infectious diseases were the fearsome killer to modern times when non-communicable diseases predominate: The following example provides an excellent way to measure sulfa’s effect.

Sulfa was not patented, appeared suddenly, and was distributed rapidly. These facts made the case ideal for study by health economists Jayachandran, Lleras-Muney, and Smith (2010). They tested population health effects of sulfa by comparing the time patterns of those mortality rates of diseases treatable by sulfa with those that weren’t. In most of these cases, the results were dramatic. Sulfa caused sharp reductions in mortality for treatable diseases such as MMR (measles, mumps and rubella), pneumonia, scarlet fever and maternal mortality. The drug lowered maternal mortality from 60 percent to 36 percent and caused similar or higher drops in the other sulfa-treatable diseases. The most telling statistic is that, by itself, sulfa raised U.S. life expectancy by 0.4 to 0.7 years.

### Prenatal Care

The importance of examining population subgroups comes from neonate mortality studies (for example, see Corman and Grossman, 1985; and Corman, Joyce, and Grossman, 1987). A neonate is an infant one month old or younger. Thus, the neonate mortality rate refers to deaths to neonates per 1,000 live births. How can we reduce these deaths? The early studies observed counties in the United States and identified several factors that seemed to cause higher neonate mortality rates. Table 5-3 displays the production of neonate mortality study by Corman, Joyce, and Grossman (1987).

The table presents their estimates of the contribution of each measured factor to the reported mortality rate decline. The WIC program is the shortened name for the governmental program designed to provide improved nutrition for women, infants, and children; it is a means-tested program, meaning that it is directed to the poor. The BCHS variable is the authors' measure combining various Bureau of Community Health Services projects, including maternal and infant care, as well as community health centers.

The data recorded in Table 5-3 indicate that of the total reduction in neonate mortality for whites during the period, 1.9 deaths per 1,000 live births, or 25.3 percent, can be explained by the observed factors, whereas for blacks a greater amount, 6.5 or 56.5 percent, can be explained. Blacks benefit more from health care on the margin, a finding that has been found in other studies. Interpret the table as follows: During the period studied, the WIC program resulted in a reduction of white neonate mortality rates of 0.425 deaths per 1,000 live births, while for blacks WIC reduced neonate mortality by 1.330 deaths per 1,000 live births.<sup>4</sup> Note too, that abortion tends to lower mortality, this result may reflect that many abortions were of fetuses that would not have survived infancy. Also note that prenatal care is effective in these data, especially for blacks; prenatal care shows up well in most studies.

A novel experiment (Evans and Lien, 2005) reinforces the value of prenatal care. When public transportation workers in Allegheny County, Pennsylvania, went on strike in 1992, prenatal visits

**TABLE 5-3 Contribution of Selected Factors to Reductions in Neonate Mortality Rates, 1964–1977**

| Factor                    | Whites Total Effect | Blacks Total Effect |
|---------------------------|---------------------|---------------------|
| Organized family planning | 0.084               | 0.526               |
| WIC                       | 0.425               | 1.330               |
| BCHS                      | 0.002               | 0.030               |
| Neonatal intensive care   | 0.140               | 0.534               |
| Abortion                  | 0.824               | 2.109               |
| Prenatal care             | 0.434               | 1.949               |
| Total explained reduction | 1.9                 | 6.5                 |
| Total reduction           | 7.5                 | 11.5                |
| Percentage explained      | 25.3                | 56.5                |

*Note:* Figures record estimates of the reduction in deaths per 1,000 live births predicted to have been caused by various factors.

*Source:* Reprinted from Journal of Human Resources, Hope Corman, Theodore J. Joyce, and Michael Grossman, "Birth Outcome Production Function in the United States," *Journal of Human Resources* 22 (1987): 339–360, with permission from The University of Wisconsin Press.

<sup>4</sup> Currie and Gruber (1996) showed general Medicaid eligibility to improve birth outcomes in a study across states in the United States. (This is distinguished from the particular Medicaid spending directed to prenatal care.) These authors, however, raised issues about the cost-effectiveness of improvements in Medicaid eligibility.

among the poor were sharply curtailed. The results were significant reductions in the quality of birth outcomes, especially for those affected during the early stages of pregnancy.

Aizer, Currie, and Moretti (2007) observed the value of prenatal care indirectly. They measured the effects of a law transferring some California Medicaid patients to managed care, which in this situation offered distinctively lower quality prenatal care. Significantly greater numbers of low-weight births were observed. Another study, by Conway and Kutinova (2006), finds prenatal care to be effective in reducing the probability of low-weight births.

### **The World's Pharmacies**

Note also the contribution of pharmaceutical availability in determining population health. Recent research (Shaw, Horrace, and Vogel, 2005; Miller and Frech, 2004) finds that countries in the Organization for Economic Cooperation and Development (OECD) with higher drug consumption have greater life expectancies. The magnitude of the effect is on a par with the effects of reduced cigarette consumption and increases of fruit and vegetables in the diet.

### **HOW DOES HEALTH CARE AFFECT OTHER MEASURES OF HEALTH?**

Mortality rates are part of the story, but only a part, and for many, not even the most important part. Other measures include morbidity data, disability days, work loss days, and other indicators. Research available also measures the quality of life.

The RAND Health Insurance Experiment (RHIE) is one of the largest randomly controlled economic experiments ever conducted. It was designed to test the effect of alternative health insurance policies on the demand for health care and on the health status of a large and closely observed group of people from all walks of life.

RAND researchers discovered that the greater the portion of the health care bill that individuals are required to pay, the less health care they choose to purchase. While this should not have been surprising, what did surprise most health economists was how great the difference was; the fully insured purchased roughly 40 percent more health care than those who had to pay their own bills. This provides an opportunity to ask whether those with 40 percent more health care were also 40 percent more healthy. This was serendipitous from a research standpoint, but a perfectly valid way to test the real contribution of health care to people's health within the context of a scientifically controlled experiment.

Fortunately, RHIE analysts kept detailed records on each person, including a dozen or more measurements under each category of physical health, mental health, social health, and general health index. They also examined their subjects' dental health, persistence of symptoms, health habits, and disability days. The results are easy to summarize. For dozens of items, virtually no differences were found between the groups studied; health care and health insurance did not seem to matter.

A simple example from the RHIE illustrates the point. Table 5-4 provides detail on work-loss days per employed person per year—a measure of health status and morbidity that some economic researchers like to use because it ties directly to both health and productivity. This table separates the RAND subjects into four groups, which differ by type of health insurance policy. Some subjects pay nothing out of pocket for their health care/health insurance package; some pay 25 percent to 50 percent of their bill themselves; others pay all of their health care bills up to a certain amount, called a deductible. The subject's out-of-pocket cost ranges from zero (free) to about 95 percent of the bill. Newhouse et al. (1993) summarize: "Our results show that the 40 percent increase in services on the free-care plan had little or no effect on health status for the average adult."

The effects on children showed a somewhat similar pattern. Valdez et al. (1985) examined data for 1,844 children in the RAND study—children who differed primarily by the type of insurance plan

**TABLE 5-4 Work Loss Days per Employed Person per Year, by Plan**

| Plan                    | Mean | Standard Error of Mean | 95% Confidence Interval | Number of Persons |
|-------------------------|------|------------------------|-------------------------|-------------------|
| Free                    | 5.47 | 0.42                   | 4.65–6.29               | 1,136             |
| Intermediate (25%, 50%) | 4.82 | 0.37                   | 4.09–5.55               | 983               |
| Individual Deductible   | 4.54 | 0.36                   | 3.83–5.25               | 787               |
| Family Deductible (95%) | 4.82 | 0.53                   | 3.78–5.86               | 600               |

Source: Reprinted by permission of the publisher from *Free for All? Lessons from the RAND Health Insurance Experiment* by Joseph P. Newhouse et al., Cambridge, MA: Harvard University Press, 1993. Copyright © 1993 by the RAND Corporation.

their families obtained. Children under the cost-sharing plans consumed up to one-third less care. However, the reduction in care was not significantly related to health status measures.

It may seem from the RAND results that public provision of health insurance to both adults and children might not be justifiable on the basis of benefits to health. However, as Jonathan Gruber (2008) points out, this conclusion does not follow. No one in the RAND Experiment was “uninsured,” completely without insurance, as are close to 50 million Americans as of this writing (the Patient Protection and Affordable Care Act begins to address them in 2014). The least insured individuals studied by RAND had full coverage for health expenditures above a deductible, which was \$1,000. Studies of the truly uninsured began to appear showing significant health gains from the provision of public insurance (Currie and Gruber, 1996; Doyle, 2005; Hanratty, 1996). These studies report reductions in infant and neonate deaths of around 5 to 10 percent. The Institute of Medicine estimates suggest that even larger gains are possible; they claim that the uninsured face a 25 percent greater mortality risks.

Gruber further explains why these studies do not conflict with RAND. He proposes that the marginal effectiveness of medical expenditures is quite high for the first expenditures but then drops off precipitously, a plausible pattern given diminishing marginal returns.

Eventually, additional spending does no good and the effectiveness curve flattens out . . . This appears to be the case as we move from less to more generous coverage, as in the RAND Health Insurance Experiment (Gruber, 2008: 584).

### On the Importance of Lifestyle and Environment

Didn't we always know that much of our health depends on the wisdom of our own choices? The role of lifestyle was best illustrated by Victor Fuchs in his book *Who Shall Live?* (1995). He compared average death rates in Nevada and Utah for 1959 to 1961 and 1966 to 1968. These two states are contiguous, and they share “about the same levels of income and medical care and are alike in many other respects” (p. 52). Nevertheless, average death rates in Nevada were greater than those in Utah. Table 5-5 shows the results of Fuchs's work. Fuchs argued that the explanation for these substantial differences surely lies in lifestyle:

Utah was, and remains, inhabited primarily by Mormons, whose influence is strong throughout the state. Devout Mormons do not use tobacco or alcohol and in general lead stable, quiet lives. Nevada, on the other hand, is a state with high rates of cigarette and alcohol consumption and very high indexes of marital and geographical instability. (p. 53)

In 2009, Utah, with its low age-adjusted death rates, was still a national leader in health (this death rate equaled 507.8), while Hawaii (717.9), and Nevada (727.3) were much higher, but



**TABLE 5-5 Excess of Death Rates in Nevada Compared with Utah, Average for 1959–1961 and 1966–1968**

| Age Group   | Males (%) | Females (%) |
|-------------|-----------|-------------|
| Less than 1 | 42        | 35          |
| 1–19        | 16        | 26          |
| 20–39       | 44        | 42          |
| 40–49       | 54        | 69          |
| 50–59       | 38        | 28          |
| 60–69       | 26        | 17          |
| 70–79       | 20        | 6           |

Source: Reprinted from Victor R. Fuchs, *Who Shall Live? Health, Economics, and Social Choice*, Expanded Edition, Singapore: World Scientific Publishing Company, 1995, p. 52, with permission from the author and World Scientific Publishing.

significantly lower than the national average (793.7). Before concluding that a simple life and plenty of sun are the tickets to good health in and of themselves, consider that many of the top 10 healthful states, while they may be sunny, are known to be chilly: Minnesota (718.6), New Hampshire (761.6), Idaho (774.5), and Colorado (620.3). (Data source: CDC, Preliminary death rates, 2009.)

### Cigarettes, Exercise, and a Good Night's Sleep

Many have chosen to quit smoking (or to avoid becoming addicted to cigarette smoking in the first place). Americans know that heart disease and cancer are the two leading killers, but most do not realize how substantial a part smoking plays. Using the category “malignant neoplasms of the respiratory system” (the category for lung cancer), we find that the 2002 death rate (51.5) is twice as high as that for any of the following: breast cancer (13.4), prostate cancer (9.2), pneumonia and influenza (17.5), diabetes mellitus (22.3), HIV (3.1), or motor vehicle–related injuries (11.8).<sup>5</sup> We already have seen the negative health production elasticity of cigarettes, which makes it clear that cigarette smoking affects the average health of the community and is statistically significant at that level.

However, economics searches for underlying causes, and human behavior can have many interwoven causes. For example, smoking and other lifestyle behaviors may themselves be determined by unobserved variables that affect health status. This common problem in economic empirical work has been addressed in recent research (Balia and Jones, 2008; Contoyannis and Jones, 2004). They address the problem by estimating both the determinants of lifestyle behaviors as well as the determinants of health status, giving a clearer picture of the importance of lifestyle. The authors showed that a good night's sleep, avoiding smoking, and regular exercise each contribute importantly to self-reported health.

While smoking certainly causes ill health, it is pleasurable as well, and one's degree of health can affect the decision to quit. For example, a healthy individual may be more likely to quit as a preventive measure; on the other hand, a critically ill individual may quit as a curative measure (Jones, 1996). Folland (2006) shows that greater life satisfaction means being less willing to risk death by smoking.

<sup>5</sup> These death rates are age-adjusted deaths per 100,000 resident population, National Vital Statistics System, 2009.

Granted that lifestyle is a major player in health comparisons between individuals, it is natural to ask whether it plays the same role when comparing countries. As we have seen earlier in the course, life expectancy in America is lower than in many developed countries. Commanor, Frech, and Miller (2006) investigated this question. They began by assessing U.S. efficiency in the production of health, finding it to be somewhat less efficient than other developed countries. What is most relevant to our present discussion is their finding that much of the U.S. deficit stems from the higher rates of obesity in the United States.

### **The Family as Producer of Health**

Women have long been warned to avoid cigarettes and alcohol while pregnant. Are such lifestyle factors important enough to be included as factors in the production of newborn health? The answer is yes. In the production of newborn birth weight (an important guide to infant health outcomes), maternal cigarette smoking has a significant negative effect (Rosenzweig and Schultz, 1983; Rosenzweig and Wolpin, 1995). Data on maternal smoking now show that taxing cigarettes leads to improved birth outcomes via its effect on smoking behaviors of expectant mothers (Evans and Ringel, 1999).

Maternal behavior also can have strong and tragic consequences in the case of drug use. Joyce, Racine, and Mocan (1992) found that the alarming increase in low-birth-weight births in New York City, particularly among blacks, was due in large part to an epidemic of illicit substance abuse by pregnant women. The explosion of cocaine use had horrible consequences for these babies.

Looking at this at a more abstract level, a study from Sweden (Bolin, Jacobsen, and Lindgren, 2002) develops the theory of how parents make health investments in themselves and their children. If parents individually make these health investment decisions strategically—that is, in response to the expected decisions of the others—the decisions, together, will not be optimal for the family. Even more significant health investment problems will occur, they warn, when parents split up in divorce, because the non-caregiver may lose some incentives to invest in the child's health. The parent's incentive to invest in the children's health is clearly a critical factor in child health.

### **Social Capital and Health**

Recent research has made it clear that family, friends, and community are associated with the health of the individual and the community. The networks of social contacts of an individual or the complex overlapping networks in a community have come to be called *social capital*. The effects, first described by political scientists, sociologists, medical researchers, and epidemiologists, suggest that social capital beneficially affects measures of health (see Islam et al, 2006, for a review).

Social capital may improve an individual's health in several ways: (1) it may relieve stress to have the support of more social contacts; (2) more contacts can provide additional information on healthful behaviors and health purchases; and (3) satisfying social relationships may provide reasons to re-evaluate risky health behaviors. This issue presents complex research obstacles; for example, not all social contacts are beneficial.

The bigger issue, however, is how to determine whether social capital in these studies causes better health or alternatively whether it is a result of some other factors. This is an important avenue by which economics and its econometric tools provide benefits to the ongoing research of other disciplines.

Health economists have taken interest in this area as a potential subject area in which to make a joint contribution with other disciplines. This is occurring both in theory, and in empirical work.<sup>6</sup> Findings generally support the hypothesis that social capital improvements lead to health improvements.

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<sup>6</sup> Folland (2006, 2008), Folland, Kaarboe, and Islam (2011).

## Environmental Pollution

Pollution causes ill health and death in individuals, with the elderly and people with respiratory diseases more susceptible. The degree to which reductions in pollution will improve the health of populations is somewhat less clear. Pollution effects on health are sizable and statistically significant in both industrialized and lesser-developed countries (Cropper et al., 1997). Based on levels of total suspended particulates (TSP) in New Delhi between 1991 and 1994, the average pollution level was five times the limit recommended by the World Health Organization (WHO). Variations in deaths in New Delhi responded statistically to the variations in pollution; if these estimates prove true, then a reduction of pollution levels of about one-third would reduce deaths by more than 2 percent.

A similar study by Schwartz and Dockery (1992) in Philadelphia suggests that reducing the pollution level there by the same 100 micrograms per cubic meter would reduce deaths by more than 6 percent in the general population and nearly 10 percent for the elderly. This is because with our generally better health status in the United States, more people live long enough to become part of the population most sensitive to respiratory problems from pollution.

## Income and Health

While we know that good health during the years when an individual is forming a career can be a big boost to that person's income later in life (James Smith, 1998), we also know that being rich does not necessarily cause one to choose to live and eat wisely. Even programs designed to raise the income of poor families, such as (the late twentieth century) Aid to Families with Dependent Children in the United States, did not always correlate with good health habits among the recipients (Currie and Cole, 1993; Currie and Gruber, 1996).

Though earlier work had suggested that being richer in America was generally better for one's health, research by Deaton and Paxson (2001) brought that conclusion into question. Examining in detail both U.S. and British data over time, they find the relation of income and health to be complex and contradictory. There was a substantial decline in mortality after 1950, but rather than growing incomes as the cause, they conclude "a more plausible account of the data is that, over time, declines in mortality are driven by technological advances, or the emergence of new infectious diseases, such as AIDS" (p. 29).

Part of our problem thus far in researching the contribution of income to health in the industrialized world is that incomes do not vary greatly enough to detect the larger patterns. Pritchett and Summers (1996) leave little doubt that extremely low incomes have a strong effect on people's health. Though they treated the econometric challenges with great respect in establishing their conclusion, the most persuasive arguments may be those provided by simple graphs fitting various health statistics to per capita income data. These curves fit well and reveal that "modern" standards of good health are enjoyed solely by the industrialized countries with mortality experience turning sharply worse with lower income levels, conditions common in the underdeveloped world.

## THE ROLE OF SCHOOLING

What is the role of education? Since education includes both formal and informal training (such as experience or on-the-job training), some portion of education is impossible to measure accurately. Most often health economists focus on schooling as measured by years of schooling, or academic accomplishments such as diplomas or degrees. Health status correlates significantly with schooling as we have seen. If the marginal product of health care is truly small ("we are on the flat of the curve"), then perhaps we should reduce public health expenditures on health care at the margin and transfer the expenditures to education. However, the wisdom of such a policy depends on which of the two theories is correct.

## Two Theories About the Role of Schooling

The ideas and work of two health economists serve to develop and contrast the two theories. First, Michael Grossman's (1972a, 1972b) theory of demand entails a central role for education. Grossman contends that better-educated persons tend to be economically more efficient producers of health status.

In contrast, Victor Fuchs (1982) has suggested that people who seek out additional education tend to be those with lower discount rates. A decision-maker with a high discount rate will tend to prefer projects with immediate payoffs versus long-term projects. People with a lower discount rate tend to be those who value the long-term gains more. Now consider individuals facing a possible investment in education. Because education requires current costs to gain distant payoffs, individuals with relatively low discount rates will be more likely to invest in education and in health as well.

## Empirical Studies on the Role of Schooling in Health

Recent evidence supports the view that education makes one a more efficient producer of health (Lleras-Muney, 2005). Knowing that compulsory education laws came into being in various places at various times in the twentieth century, she reasoned that the related birth cohorts from that era would have experienced different levels of education but would have been similar in many other respects. This formed a natural experiment in which she could analyze the survival patterns of these people to detect a pure influence of education on health. Furthermore, the education laws could not have been directly manipulated by the study subjects, so they were good "instruments" for education. By this approach, she was able to conclude that education has a clear, causal, and positive effect on health. By 1960, the early century education experience appeared to have increased life years by 1.7 years, a substantial increase and one not due to time preferences of the subjects.

Lleras-Muney's study inspired new research of the effects of new laws extending the length of compulsory education in England and Ireland (Oreopoulos, 2006; Auld and Sidhu, 2005). These supported the earlier findings; an additional year of schooling caused an improvement in the affected student's health. To emphasize, the improved health was experienced by "likely dropouts," forced by the law to attend one more year of high school. Interestingly, Lindeboom and colleagues (2006) inquired through research as to whether the children born later on to these students *also* benefited from improved health, but the findings indicated that they did not.

In summary, research has supported the theory that education makes people more efficient producers of their own health. Cutler and Lleras-Muney (2006) add further support in their recent work by showing that education is associated statistically with better reasoned choices of health related behaviors. One finds as well that education plays a stronger role in health for cases where new medical knowledge is more important.

## CONCLUSIONS

In this chapter, we investigated many topics related both directly and indirectly to the production of health. The health production function exhibits the law of diminishing marginal returns. While the total contribution of health care is substantial, the marginal product is often small. Historically, we found that much of the decline in mortality rates occurred prior to the introduction of specific, effective medical interventions. Thus, historically the contributions of health care, at least as provided by the health practitioner, were probably small until well into the twentieth century. The small, modern-day marginal product of health care is statistically significant. Health care benefits people differentially and is generally more productive on the margin for women and blacks. Similarly, certain categories of health care have greater marginal effects on the population than others; prenatal care programs are examples of the more productive categories. Education has a strong association with health status. Whether this means that it causally improves health has long been an issue of contention. Recent research supports the view that education improves health.

## Summary

1. The production function for health exhibits diminishing marginal returns. In developed countries, the total product of health care is probably substantial at the same time that the marginal product is relatively small.
2. The historical declines in mortality rates in representative industrial countries were substantially responsible for the large growth of populations.
3. The historical declines in population mortality rates were not due to medical interventions because effective medical interventions became available to populations largely after the mortality had declined. Instead, public health, improved environment, and improved nutrition probably played substantial roles.
4. The marginal product of health care in the United States is small. Recent studies find elasticities in a narrow range around 0.10. Nevertheless, the total contribution of investment in health care technology over the past several decades is probably in the trillions of dollars; much of the improvement in life expectancy in this period can be attributed to health care improvements.
5. The RAND Health Insurance Experiment found that increased use of health care has little effect on the illness rates of the study population. However, studies of the totally uninsured now reveal gains in health due to publicly provided health insurance.
6. Lifestyle and environment are major and statistically significant determinants of population health status.
7. Health care contributes more substantially to health for subgroups of the population, including infants and also certain ethnic minority groups.
8. Social capital, produced by groups, is increasingly viewed as a substantive determinant of individuals' health.
9. Education, as measured by years of schooling, is positively related to population health. Recent research supports the view that the relation is causal, that increased education improves health.

## Discussion Questions

1. Assume that health production is subject to diminishing returns and that each unit of health care employed entails a constant rate of iatrogenic (medically caused) disease. Would the production of health function eventually bend downward? Explain.
2. What evidence is there to suggest that the United States is on the "flat of the curve" in health production? Is a typical developing country likely to be on the flat of its health production function? Discuss the differences.
3. Which of the following are important in explaining the modern rise in population in England and Wales: birthrates, death rates, and net migration rates? Describe the evidence.
4. "Medical interventions were not important in the historical declines in mortality rates, but that does not imply that medical research was unimportant." Explain this viewpoint.
5. What role did public health play in the historical decline in mortality rates?
6. Suppose you were hired as an adviser to a developing country and you were versed in the theory of production, the historical role of medicine, and the modern-day health production function studies. Their government seeks advice on the wisdom of a relative emphasis on health and health investment versus other forms of economic investment. What would be your advice?
7. Someone says the following: "Lifestyle may be the most important determinant of health status, but changing lifestyles may not be the least costly way to improve population health status." Explain the circumstances under which this opinion could be true. Is it likely to be true in reality? What does the evidence on lifestyle suggest about government policies to improve the public's overall health?
8. Summarize the two theories on how schooling is correlated with health status. Which of the two theories does the evidence support?
9. Research shows that the returns for prenatal health care are high, whereas it may cost hundreds of thousands of dollars to keep an acutely ill, elderly person alive. What does this suggest about the appropriate allocation of resources among members of society?
10. What is the total contribution of health care to health as estimated by researchers? What parts of health care are most effective in this regard?

## Exercises

1. Graph the production of health function  $HS = 10HC^{0.5}E^{0.3}LS^{0.4}HB^{0.2}$  in a graph with axes  $HS$  and  $HC$ , assuming  $E = 10$ ,  $LS = 5$ , and  $HB = 7$ . Graph the marginal product of health inputs. Is it increasing or decreasing? Show how the curve changes when  $E$  is increased to 15.
2. Which factors in Table 5-3 were important in explaining improvements in black neonate mortality rates? White neonate mortality rates? Speculate on why some of these factors may have been more important for blacks.
3. What are the differences between mortality and morbidity? Would you expect the two variables to be related to each other? If so, how?
4. We know that correlations never explain; it is our theories that provide explanations. Reexamine Table 5-3 and draft theories to explain why WIC has a larger contribution than Organized Family Planning. Why do the contributions differ between blacks and whites on WIC, abortion, and pre-natal care?
5. Pritchett and Summers argue that income per capita is strongly and positively related to health status when viewed across the world. From data in Table 2-2, in Chapter 2 of this text, plot a graph of GDP per capita against life expectancy for the countries shown. Does your plot confirm the Pritchett and Summers finding?